

I-VOTING (ESTONIA)



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Estonia and it's general facts:

- Estonia is the country located in northern Europe.
- Capital city: Tallin
- Official language : Estonian
- Estonia has one of the **highest startup rates per capita** in Europe.
- More than **50% of all votes** in recent elections are cast online
- Estonia pioneered the **e-Residency program**,
 - ✓ has global success in startups like Skype, Bolt. Wise(transfer wise and Pipedrive)
 - ✓ has a **paperless cabinet meeting** system,
 - ✓ uses **blockchain technology** in government systems for data integrity.

What is I voting process

- I-voting is the system that allows Estonian citizens to **vote online from anywhere** at home, abroad, or on any internet-connected device during the **advance voting period** of an election. Voters authenticate themselves using secure digital identity instruments like an **ID card**, **Mobile-ID**, or **Smart-ID**.
- **Inclusive:** From 2017, 16- and 17-year-olds can use I-voting in local elections.
- Estonia introduced i-Voting in 2005, becoming the first country in the world to hold nationwide online elections.

How It Works ?

Start the advance voting period: I-voting opens on Monday of election week and closes Saturday evening.

Authenticate: Voters log in using their Estonian digital ID (ID-card, Mobile-ID, or Smart-ID).

Select a candidate: The i-voting application shows eligible candidates.

Encrypt and sign: The vote is encrypted and digitally signed by the voter — ensuring both secrecy and authenticity.

Send and verify: Voters can later verify that their cast vote reached the system using a separate app.

Vote Changes and Final Count

- Re-voting:** Voters may submit a new i-vote as many times as they want during the period. only the **last one counts**.
- Paper override:** If a voter casts a **physical ballot** at a polling station, their i-vote is cancelled.
- Counting:** After election day, i-votes are anonymised and decrypted only for tallying by the National Electoral Committee

Security and Oversight

Estonia's authorities emphasize that i-voting is **secure and transparent**:

Encryption and anonymity: Votes are encrypted and later anonymised before counting.

Audits and observers: Independent audits and observers review procedures for each election.

Verification: Voters can verify (within ~15 minutes) that their vote was correctly received.

Legal framework: The system operates under Estonia's election laws and court rulings on equal treatment of voters.

Estonia has experienced **major cyber attacks**, the most famous being the **2007 nationwide cyber attack**, often described as the **first large-scale cyber war attack** on a country.

- Later on, Estonia strengthened its cybersecurity infrastructure.
- NATO established the **Cooperative Cyber Defence Centre of Excellence (CCDCOE)** in Tallinn in 2008.
- Estonia became a **global leader in cyber resilience**, digital ID security, and online election protection

How i-Voting Has Helped Estonia's Public Service Management (PSM)

1.Increased administrative efficiency by reducing the burden of manual polling, paperwork, and physical logistics.

2.Lower operational costs for elections because fewer polling stations, staff, and materials are needed.

3.Faster election processes, including quicker vote counting and result verification.

4.Improved service accessibility, especially for citizens living abroad, rural populations, the elderly, and people with disabilities.

5.Enhanced data accuracy and record-keeping, reducing human error in voter lists and vote management.

6.Strengthened citizen–state interaction, making elections a modern, user-friendly public service.

THANK YOU !!
ANY QUERIES ?